Quantum Field Theory

Exercise 5.1

$$\mathcal{I} = \int d^4k \, \frac{1}{k^2} \, \frac{1}{(k+p)^2 - m^2}$$

Feynman parametrization: $$\frac{1}{AB} = \int_0^1 dx \, \frac{1}{(Ax + B(1-x))^2}$$

Let's use that for $\mathcal{I}$:

$$\mathcal{I} = \int d^4k \int_0^1 \frac{dx}{\left[\left((k+p)^2 - m^2\right)x + k^2(1-x)\right]^2}$$

$$\mathcal{I} = \int d^4k \int dx \left[k^2x + p^2x + 2kpx - m^2x + k^2 - k^2x \right]^{-2}$$

$$= \int d^4k \int dx \left[p^2x + 2k\cdot px - m^2x + k^2 \right]^{-2} = \int d^4k \int dx \left[\underbrace{(k+px)^2}_{k\text{-dependent part}} - p^2x^2 - m^2x + p^2x \right]^{-2}$$

We have isolated the k-dependent part with other terms (both are still dependent on x)

$\Rightarrow$ Let's shift the integration variable: $l = k + px \Rightarrow \frac{dk}{dl} = 1 \Rightarrow dl = dk$

and let's just write $\Delta^2 \equiv p^2x - m^2x - p^2x^2$

$$\Rightarrow \quad \mathcal{I} = \int dx \int d^4l \, (l^2 - \Delta^2)^{-2}$$

We now Wick-rotate to get from Minkowski space to Euclidean space with the help of Cauchy's theorem: variable chang from $l \to l_E$ with $l^0 = i l_E$, $l^i = l^i_E$

$$\Rightarrow \quad \left[\int_{-\infty}^{+\infty} + \int_{+i\infty}^{-i\infty} \right] dl^0 \, (l^2 - \Delta^2)^{-2} = 0 \qquad \Big\| \text{ Cauchy's theorem.}$$

$$\Rightarrow \quad \int_{-\infty}^{+\infty} dl^0 \, (l^2 - \Delta^2)^{-2} = - \int_{+i\infty}^{-i\infty} dl^0 \, (l^2 - \Delta^2)^{-2}$$

$$= -i \int_{\infty}^{-\infty} dl^0_E \, (-l^{0\,2}_E - l^{i\,2}_E - \Delta^2)^{-2}$$

$$= i \int_{-\infty}^{\infty} dt \, (-t^2 - r^2 - \Delta^2)^{-2}$$

$$\int_{-\infty}^{+\infty} dl^0 \, (l^2 - \Delta^2)^{-2} = i \int_{-\infty}^{\infty} dt \left[t^2 + r^2 + \Delta^2 \right]^{-2}$$

Now here we use the $l^0 = i l^0_E$, $l^i = l^i_E \Rightarrow dl^0 = i \, dl_E$

limit of integration

$-i\infty = i l_E \Rightarrow l_E \to -\infty$

$i\infty = i l_E \Rightarrow l_E \to \infty$

let's write $l^0_E = t$

and $l^{i\,2}_E = r^i \quad \Rightarrow \quad t^2 + r^2 = l_E^2$

$(-1)^{-2} = 1$

So l_E is Euclidean 4-dimensional vector

Now can write Minkowski d^4l integral in-terms of Euclidean integral $d^4 l_E$

$$\Rightarrow \int d^4l \, (l^2 - \Delta^2)^{-2} = i \int d^4 l_E \left[l_E^2 + \Delta^2 \right]^{-2}$$

To evaluate this integral we can use the spherical coordinates :

$$\int d^4 l_E = \int d\Omega_4 \int |l_E|^3 \, d|l_E| \qquad \text{with} \quad \int d\Omega_4 = 2\pi^2$$

$$\frac{d}{d\xi} \frac{\xi^2}{(\xi^2+\Delta^2)} = -\frac{2\xi^3}{(\xi^2+\Delta^2)^2} + \frac{2\xi}{\xi^2+\Delta^2}$$

$$\frac{d}{d\xi} \ln(\xi^2+\Delta^2) = \frac{2\xi}{\xi^2+\Delta^2}$$

so

$$\frac{1}{2}\frac{d}{d\xi}\left[-\frac{\xi^2}{\xi^2+\Delta^2} + \ln(\xi^2+\Delta^2)\right] = \frac{\xi^3}{(\xi^2+\Delta^2)^2}$$

$$\Rightarrow I = i2\pi^2 \int dx \int d|l_E| \, |l_E|^3 \left(|l_E|^2 + \Delta^2 \right)^{-2}$$

$$= 2\pi^2 i \int dx \int_0^\infty d|l_E| \frac{|l_E|^3}{(|l_E|^2 + \Delta^2)^2}$$

$$= 2\pi^2 i \int dx \int_0^\infty d|l_E| \frac{d}{d|l_E|}\left\{ -\frac{|l_E|^2}{|l_E|^2+\Delta^2} + \ln(|l_E|^2+\Delta^2) \right\}$$

$$= 2\pi^2 i \int dx \left\{ -\frac{|l_E|^2}{|l_E|^2+\Delta^2} + \ln(|l_E|^2+\Delta^2) \right\} \Bigg|_{|l_E|=0}^{|l_E|=\infty}$$

This has divergence at $|l_E| \to \infty$ so let's write in terms of $|l_E|$ as upper limit

$$= -2\pi^2 i \int dx \left[\ln(\Delta^2) + \frac{|l_E|^2}{|l_E|^2+\Delta^2} + \ln(|l_E|^2 + \Delta^2) \right]$$

↑ diverges as $|l_E| \to \infty$

Since we encounter divergence at $|l_E| \to \infty$ limit we just calculated the integral without taking the limit to ∞ (since it diverges). We will just write this as the integral but it should be noted that there is still limit $|l_E| \to \infty$ to be taken. Essentially we have introduced a cut off at some $|l_E|$.

Now let's regularize the incountered divergence via Pauli-Villars regularization : This amounts to $\frac{1}{k^2} \to \frac{1}{k^2} - \frac{1}{k^2 - \Lambda}$, where Λ is large

Now we have a integral $I_{pv} = \int d^4k \left[\frac{1}{k^2} - \frac{1}{k^2 - \Lambda} \right] \frac{1}{(k+p)^2 - m^2}$

$$I_{pv} = \underbrace{\int d^4k \frac{1}{k^2}\frac{1}{(k+p)^2 - m^2}}_{= I} - \underbrace{\int d^4k \frac{1}{k^2 - \Lambda^2}\frac{1}{(k+p)^2 - m^2}}_{\equiv J}$$

We have already calculated I so let's now calculate J

$$J = \int d^4k \, \frac{1}{k^2 - \Delta_\Lambda^2} \frac{1}{(k+p)^2 - m^2}$$

// Feynman parametrization: $\frac{1}{AB} \int_0^1 dx \, \frac{1}{(Ax + (1-x)B)^2}$

in our case: $A = (k+p)^2 - m^2$ and $B = k^2 - \Delta_\Lambda^2$

$$= \int dx \int d^4k \left[(k+p)^2 x - m^2 x + k^2 - \Delta_\Lambda^2 - k^2 x + \Delta_\Lambda^2 x \right]^{-2} = \int dx \int d^4k \left[\cancel{k^2 x} + p^2 x + 2kpx - m^2 x - \cancel{k^2 x} + k^2 + \Delta_\Lambda^2 x - \Delta_\Lambda^2 \right]^{-2}$$

$$(k+px)^2 = 2kpx + k^2 + p^2 x^2 \Rightarrow p^2 x + 2kpx - m^2 x + k^2 + \Delta_\Lambda^2 x - \Delta_\Lambda^2 = (k+px)^2 \underbrace{- p^2 x^2 + p^2 x - m^2 x + \Delta_\Lambda^2 x - \Delta_\Lambda^2}_{= \Delta_\Delta^2}$$

(doesn't depend on k)

$$\Rightarrow J = \int dx \int d^4k \left[(k+px)^2 - \Delta_\Delta^2 \right]^{-2}$$

// let's do a shift: $l = k + px$ $\Rightarrow dk = dl$

$$\Rightarrow J = \int dx \int d^4l \, (l^2 - \Delta_\Delta^2)^{-2}$$

, Now we Wick-rotate like previously with

$l^0 = i l_E^0$, $l^i = l_E^i$ $\Rightarrow dl^0 = i \, dl_E$ to get from Minkowski space to Euclidean space.

$$\Rightarrow \int_{-\infty}^{\infty} dl^0 \, (l^2 - \Delta_\Delta^2)^{-2} = i \int_{-\infty}^{\infty} dl_E^0 \left[l_E^2 + \Delta_\Delta^2 \right]^{-2}$$

// with $l_E^2 = l_E^{0\,2} + l_E^{i\,2}$

This gives us $J = i \int dx \int d^4 l_E \left[l_E^2 + \Delta_\Delta^2 \right]^{-2}$

// We can use the spherical coordinates with $\int d\Omega_4 = 2\pi^2$

$$J = 2\pi^2 i \int dx \int d|l_E| \, \frac{|l_E|^3}{(|l_E|^2 + \Delta_\Delta)^2}$$

// This is exactly same integral as I but with Δ_Δ instead of Δ which does not effect the $d|l_E|$ integral.

$$\Rightarrow J = 2\pi^2 i \int dx \left\{ - \frac{|l_E|^2}{|l_E|^2 + \Delta_\Delta^2} + \ln\left(|l_E|^2 + \Delta_\Delta^2 \right) \right\} \Bigg|_{|l_E| = 0}^{|l_E| = \infty}$$

We once again only write/calculate the $|l_E| = 0$ limit and leave $|l_E| = \infty$ diverging limit implicitely and get

$$J = -2\pi^2 i \int dx \left[\ln(\Delta_\Delta^2) + \frac{|l_E|^2}{|l_E|^2 + \Delta_\Delta^2} + \ln\left(|l_E|^2 + \Delta_\Delta^2 \right) \right]$$

Now we can subtract this from I to get I_{pv}

$$I_{pv} = I - J = -2\pi^2 i \int dx \left\{ \ln(\Delta^2) - \ln(\Delta_\Delta^2) + \frac{|l_E|^2}{|l_E|^2 + \Delta^2} - \frac{|l_E|^2}{|l_E|^2 + \Delta_\Delta^2} + \underbrace{\ln\left[|l_E|^2 + \Delta^2 \right] - \ln\left[|l_E|^2 + \Delta_\Delta^2 \right]}_{= \ln\left[\frac{|l_E|^2 + \Delta^2}{|l_E|^2 - \Delta_\Delta^2} \right]} \right\}$$

Since there is implicit limit of $|l_E| \to \infty$ we now take the limit and the terms of form $\frac{|l_E|^2 \pm u}{|l_E|^2 \pm v} \to 1$ as $|l_E| \to \infty$ and $\ln(1) = 0$

$$\Rightarrow I_{pv} = -2\pi^2 i \int dx \, \ln\left[\frac{\Delta^2}{\Delta_\Delta^2} \right] = 2\pi^2 i \int dx \, \ln\left[\frac{\Delta_\Delta^2}{\Delta^2} \right]$$

This is the logarithmic divergence we were looking for.

Exercise 5.2

Compute

$$\int d\Omega_k \left[\frac{2p\cdot p'}{(p\cdot k)(p'\cdot k)} - \frac{m^2}{(p\cdot k)^2} - \frac{m^2}{(p'\cdot k)^2} \right]$$

Let's work in a frame where $\vec{p}\,' = -\vec{p} \Rightarrow p^2 = p'^2 = m^2$ since p' and p is momentum of same particle (electron). $q^2 = (p-p')^2 \Rightarrow -(2\vec{p})^2 = -4|\vec{p}|^2$ in the chosen frame.

$$p\cdot k = E\cdot k^0 - \vec{p}\cdot\vec{k}$$

$$E = \sqrt{m^2 + |\vec{p}|^2} = |\vec{p}|\sqrt{\frac{m^2}{|\vec{p}|^2} + 1}$$

$\beta = m^2/(-q^2)$

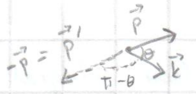
$$\frac{m^2}{|\vec{p}|^2} = \frac{-4m^2}{-4|\vec{p}|^2} = \frac{4m^2}{-q^2} = 4\beta$$

$E = |\vec{P}|\sqrt{4\beta+1} = |\vec{P}|\sqrt{1+4\beta}$ and because $k^2 = 0$ we have $k^0 = |\vec{k}|$

$\Rightarrow \quad E\cdot k^0 = |\vec{P}||\vec{k}|\sqrt{1+4\beta}$ Now let's define the angle between $\vec{P}$ and $\vec{k}$ to same as the angle θ in Ω_k by setting the coordinate to have origin at the necessary point. With this we can write

$\vec{P}\cdot\vec{k} = |\vec{P}||\vec{k}|\cos\theta$ and $\vec{P}'\cdot\vec{k} = |\vec{P}'||\vec{k}|\cos(\theta') = |\vec{P}||\vec{k}|\cos(\pi-\theta) = -|\vec{P}||\vec{k}|\cos\theta$

since



$\Rightarrow \quad p\cdot k = |\vec{P}||\vec{k}|\left(\sqrt{1+4\beta} - \cos\theta\right)$ and

$$p'\cdot k = |\vec{P}||\vec{k}|\left(\sqrt{1+4\beta} + \cos\theta\right)$$

And finally $p\cdot p' = E^2 + |\vec{P}|^2 = |\vec{P}|^2[1+4\beta+1] = |\vec{P}|^2(2+4\beta)$

Let's Now input these to the integral

$$I \equiv \int d\Omega_k \left(\frac{2p\cdot p'}{(p\cdot k)(p'\cdot k)} - \frac{m^2}{(p\cdot k)^2} - \frac{m^2}{(p'\cdot k)^2} \right) = \int d\Omega_k \left[\frac{|\vec{P}|^2(2+4\beta)\cdot 2}{|\vec{P}||\vec{k}|(\sqrt{1+4\beta}-\cos\theta)|\vec{P}||\vec{k}|(\sqrt{1+4\beta}+\cos\theta)} \right.$$

$$\left. - \frac{m^2}{|\vec{P}|^2|\vec{k}|^2(\sqrt{1+4\beta}-\cos\theta)^2} - \frac{m^2}{|\vec{P}|^2|\vec{k}|^2(\sqrt{1+4\beta}+\cos\theta)^2} \right]$$

$x^2 - y^2 = (x+y)(x-y)$

let's define

$R \equiv \sqrt{1+4\beta}$ for convinience

$\Omega_k = d\cos\theta\, d\phi$ and $\int d\phi = 2\pi$

also $\frac{m^2}{|\vec{P}|^2} = 4\beta$

$$\Rightarrow I = \frac{2\pi}{|\vec{k}|^2}\int d\cos\theta \; \frac{4(1+2\beta)}{R^2 - \cos^2\theta} - \frac{4\beta}{(R-\cos\theta)^2} - \frac{4\beta}{(R+\cos\theta)^2}$$

The limit for $\cos\theta$ is from -1 to 1

Let's first solve the first term in the integral by writing $u \equiv \cos\theta$ for convinience

$\int \frac{1}{R^2 - u^2}du = \frac{1}{(R-u)(R+u)}$, we can factorize this to be of form $\frac{a}{R-u} + \frac{b}{(R+u)}$ for some a and b

$$\frac{a}{(R-u)} + \frac{b}{(R+u)} = \frac{aR + au + bR - bu}{R^2 - u^2} = \frac{u(a-b) + R(a+b)}{R^2 - u^2}$$

This is equal to $\frac{1}{R^2-u^2}$ with $a = b = \frac{1}{2R}$

$$\Rightarrow \int_{-1}^{1} \frac{1}{R^2 - u^2}\, du = \frac{1}{2R}\int_{-1}^{1} \frac{1}{R-u} + \frac{1}{R+u}\, du = \frac{1}{2R}\Big[\underbrace{-\ln[R-u] + \ln[R+u]}_{=\ln\left(\frac{R+u}{R-u}\right)}\Big]\Big|_{u=-1}^{u=1}$$

$$= \frac{1}{2R}\left[\ln\left[\frac{R+1}{R-1}\right] - \ln\left[\frac{R-1}{R+1}\right]\right] = \underline{\frac{1}{R}\ln\left[\frac{R+1}{R-1}\right]}$$

$$= \frac{1}{\sqrt{1+4\beta}} \ln\left(\frac{\sqrt{1+4\beta}+1}{\sqrt{1+4\beta}-1}\right)$$

And the second term is integrated as (again we set $u = \cos\theta$ for convinience $\Rightarrow du = d\cos\theta$

$$\int_{-1}^{1} \frac{1}{(R-u)^2} + \frac{1}{(R+u)^2} = \frac{1}{R-u} - \frac{1}{R+u}\Big|_{u=-1}^{u=1} \qquad \Big\| \quad \frac{d}{dx}\frac{1}{R-x} = -1\cdot\frac{-1}{(R-x)^2} = \frac{1}{(R-x)^2}$$

$$= \frac{1}{R-1} - \frac{1}{R+1} - \frac{1}{R+1} + \frac{1}{R-1} = \frac{2}{R-1} - \frac{2}{R+1}$$

$$= \frac{2(R+1) - 2(R-1)}{R^2 - 1} = \frac{4}{R^2-1} = \frac{4}{1+4\beta-1} = \frac{1}{\beta}$$

Now we put these results together to get

$$I = \frac{2\pi}{|\vec{k}|^2} \cdot \int d\cos\theta \; 4(1+2\beta)\frac{1}{R^2 - \cos^2\theta} - 4\beta\left[\frac{1}{(R-\cos\theta)^2} + \frac{1}{(R+\cos\theta)^2}\right]$$

$$= \frac{2\pi}{|\vec{k}|^2}\left[\frac{4(1+2\beta)}{\sqrt{1+4\beta}} \ln\left(\frac{\sqrt{1+4\beta}+1}{\sqrt{1+4\beta}-1}\right) - \frac{4\beta}{\beta}\right]$$

$$I = \frac{8\pi}{|\vec{k}|^2}\left[\frac{1+2\beta}{\sqrt{1+4\beta}} \ln\left(\frac{\sqrt{1+4\beta}+1}{\sqrt{1+4\beta}-1}\right) - 1\right] \blacksquare$$

We have computed the angular integral!!

Exercise 5.3

Calculate: $I \equiv \int dx\,dy\,dz\; \delta(1-x-y-z) \frac{1}{-xyq^2 + (1-z)^2 m^2 + \mu^2}$

Integrate over dx: $I = \int_0^1 dz \int_0^{1-z} dy \frac{1}{-(1-y-z)yq^2 + (1-z)^2 m^2 + \mu^2}$

Change of variables: $y = (1-z)\xi \Rightarrow dy = (1-z)\,d\xi$ and $w = (1-z)$

$$\Rightarrow \frac{dw^2}{dz} = -2(1-z) \Rightarrow dz = \frac{-1}{2(1-z)}\, d(w^2) \Rightarrow y = w\xi, \Rightarrow 1-y-z = w - w\xi$$

$$\Rightarrow (1-y-z)y = \xi(1-\xi)w^2 \quad \text{and} \quad (1-z)^2 = w^2$$

$$\Rightarrow \quad I = \int_0^1 d\xi \int_1^0 \frac{-1}{2} \cdot \frac{d(\omega^2)}{-(\xi(1-\xi))\omega^2 q^2 + \omega^2 m^2 + \mu^2} \qquad \Big\| \quad \frac{m^2}{-q^2} = \beta \quad \text{and let's define } R \equiv \frac{\mu^2}{-q^2}$$

$$= \frac{1}{2}\int_0^1 d\xi \int_0^1 \frac{1}{-q^2} \, \frac{d(\omega^2)}{[\xi(1-\xi)+\beta]\omega^2 + R^2} \qquad \Big\| \quad \frac{d}{dx}\ln(ax+b) = \frac{a}{ax+b}$$

$$= \frac{-1}{2q^2}\int_0^1 d\xi \, \frac{1}{\xi(1-\xi)+\beta}\left(\ln\big[[\xi(1-\xi)+\beta]+R^2\big] - \ln\big[R^2\big]\right)$$

As $R^2 \to 0$ the term $\frac{1}{\xi(1-\xi)+\beta}\ln[\xi(1-\xi)+\beta+R^2]$ is very small since the value of $\frac{1}{\xi(1-\xi)+\beta}\ln[\xi(1-\xi)+\beta]$ is very small since $0<\xi<1$ and as $R \to 0$ the another term diverges $\Rightarrow$ the $\ln[R^2]$ is dominant and $\gg$ than $\ln[\xi(1-\xi)+\beta+R^2]$ term so we can approximate

$$I \simeq -\frac{\ln[R^{-2}]}{2q^2}\int_0^1 d\xi \, \frac{1}{\xi(1-\xi)+\beta}$$

Let's look for root of polynomial $\xi(1-\xi)+\beta$ to factorize the integrand

$\xi - \xi^2 + \beta$ has root of $\xi_0 = \frac{-1 \pm \sqrt{1+4\beta}}{-2}$

$$\Rightarrow \quad \xi - \xi^2 + \beta = -\left(\xi - \frac{1}{2} - \frac{\sqrt{1+4\beta}}{2}\right)\left(\xi - \frac{1}{2} + \frac{\sqrt{1+4\beta}}{2}\right) \qquad \Big\| \quad \text{let's write } \frac{\sqrt{1+4\beta}}{2} \equiv k$$

let's also do change of variables: $u = (\xi - \frac{1}{2}) \Rightarrow d\xi = du$

$$\Rightarrow \quad I = -\frac{\ln(R^{-2})}{2q^2}\int_{-\frac{1}{2}}^{\frac{1}{2}} du \, \frac{1}{(u-k)(u+k)} = -\frac{\ln(R^{-2})}{2q^2}\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{du}{(k-u)(k+u)} \qquad \Big\| \quad \frac{1}{k-u}\,\frac{1}{k+u} = \frac{1}{2k}\left[\frac{1}{k+u} + \frac{1}{k-u}\right]$$

$$I = -\frac{\ln(R^{-2})}{2q^2 \, 2k}\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{k+u} + \frac{1}{k-u} = -\frac{1}{2q^2}\,\frac{1}{2k}\Big[\ln(k+u) - \ln(k-u)\Big]_{u=-1/2}^{u=1/2}$$

$$= \frac{-\ln(R^{-2})}{2q^2 k}\ln\left[\frac{k+1/2}{k-1/2}\right] = \frac{1}{2q^2}\,\frac{-2}{\sqrt{1+4\beta}}\left[\frac{\frac{1}{2}(\sqrt{1+4\beta}+1)}{\frac{1}{2}(\sqrt{1+4\beta}-1)}\right] \qquad \Big\| \quad R^{-2} = \frac{-q^2}{\mu^2}$$

$$= \frac{1}{2}\,\frac{1}{q^2\sqrt{1+4\beta}}\,\ln\left(\frac{-q^2}{\mu^2}\right)\left[-2\ln\frac{\sqrt{1+4\beta}+1}{\sqrt{1+4\beta}-1}\right] \quad \blacksquare$$

We have calculated the integral